Signals and Systems — Reference

the big picture

Everything in this course is one of two things:

- a **signal**: a quantity that changes over time, like a voltage or a sound.
- a **system**: a machine that takes a signal in and puts a signal out.

The topics build on each other in this order:

1. **How big is a signal?** Energy and power (4–5).
2. **Building blocks.** The button δ, the switch u, and the ramp r (10).
3. **Machines.** Systems, block diagrams, difference equations (11).
4. **Describing machines.** Memory, causality, stability, linearity, time invariance (12).
5. **One echo describes the machine.** The impulse response h and the convolution sum $y = x * h$ (13).

The link from 1 to 4: BIBO stability asks if a *bounded* input gives a *bounded* output. That is a question about signal size.

plain-English glossary

Symbol	Say it as
$\delta[n]$, $\delta(t)$	**the button**: pressed once at time 0, then released
$u[n]$, $u(t)$	**the switch**: flipped on at time 0 and left on
$r[n]$, $r(t)$	**the ramp**: the running total of the switch, $0, 1, 2, 3, \ldots$
t	the time variable. It sweeps through every time, and is gone after you integrate
T	the half-width of a window $[-T, T]$. It is the thing that goes to ∞
t_0, n_0	"t-naught": one fixed time, such as 3
N	DT window half-width ($2N+1$ samples), or a DT period
ε	"epsilon": a tiny number that we shrink toward 0
$\int \ldots dt$	add up area as t sweeps (CT)
$\sum$	add up separate terms (DT), the DT twin of $\int$
dt	an infinitely thin slice of time. Not Δt.
B	a ceiling: the input never goes above B in size
$h[n]$	**the echo** (impulse response): what an LTI machine puts out for one button press
$x * h$	"x convolved with h: add up one echo per input stem

notation: CT and DT side by side

Item	Continuous time (CT)	Discrete time (DT)
signal	$x(t)$, round brackets	$x[n]$, square brackets
time variable	t, any real number	n, integers only
period	T (seconds)	N (samples, an integer)
frequency	ω (rad/s), f (Hz)	$\hat{\omega}$ (rad/sample)
energy, power	integral $\int$	sum $\sum$
rate of change	derivative $\frac{d}{dt}$	first difference $x[n] - x[n-1]$
running total	integral $\int_{-\infty}^{t}$	running sum $\sum_{k=-\infty}^{n}$
button	$\delta(t)$, area 1	$\delta[n]$, height 1

sinusoid quantities

Symbol	Name	Unit	Link
A	amplitude	same as x	peak value
ω	angular frequency	rad/s	$\omega = 2\pi f$
f	frequency	Hz = cycles/s	$f = \frac{1}{T}$
T	period	s	$T = \frac{2\pi}{\omega}$
ϕ	phase	rad	$\phi = -\omega t_0$
t_0	time shift	s	$t_0 = -\frac{\phi}{\omega}$
$\hat{\omega}$	DT frequency	rad/sample	$\hat{\omega} = \omega T_s$
T_s, f_s	sampling period, rate	s, Hz	$f_s = \frac{1}{T_s}$

trig and complex number facts

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

$\sin 0 = 0$ and $\cos 0 = 1$. Do not swap them.

sine as a cosine	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$
negative amplitude	$-A\cos\theta = A\cos(\theta + \pi)$
square of a cosine	$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$
cosine from exponentials	$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$
sine from exponentials	$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$
size of $a + jb$	$\sqrt{a^2 + b^2}$
angle of $a + jb$	$\arctan\frac{b}{a}$ (check the quadrant)
polar form	$C = \lvert C\rvert e^{j\phi}$
size of $e^{j\theta}$	1
whole turns	$e^{j2\pi k} = 1$ for each integer k

sums, integrals, limits

Item	Result
infinite geometric sum, $\lvert r\rvert < 1$	$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$
finite geometric sum, $r \neq 1$	$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$
count of terms from $-N$ to N	$2N+1$
$\int t^n\, dt$	$\frac{t^{n+1}}{n+1}$
$\int e^{at}\, dt$	$\frac{e^{at}}{a}$
$\int_0^{\infty} e^{-at}\, dt, \quad a > 0$	$\frac{1}{a}$
definite integral	$F(\text{top}) - F(\text{bottom})$
fixed number over a growing T	$\lim_{T\to\infty} \frac{c}{T} = 0$
a growing T alone	$\lim_{T\to\infty} cT = \infty \quad (c > 0)$
T cancels	$\frac{cT}{2T} = \frac{c}{2}$, a constant

formula sheet

1. Signals and sampling

sampling	$x[n] = x(nT_s)$
sampling rate	$f_s = \frac{1}{T_s}$

2–3. Sinusoids

CT sinusoid	$x(t) = A\cos(\omega t + \phi)$
angular frequency	$\omega = 2\pi f$
period	$T = \frac{1}{f} = \frac{2\pi}{\omega}$
phase $\to$ time shift	$t_0 = -\frac{\phi}{\omega}$
time shift $\to$ phase	$\phi = -\omega t_0$
DT sinusoid	$x[n] = A\cos(\hat{\omega} n + \phi)$
CT $\to$ DT frequency	$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$
DT $\to$ CT frequency	$f = \frac{\hat{\omega} f_s}{2\pi}$
aliases	$\pm\hat{\omega} + 2\pi k, \quad \pm f + k f_s$

4–5. Energy and power

CT energy (total)	$E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt$
DT energy (total)	$E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
CT power (average)	$P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2\,dt$
DT power (average)	$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} \lvert x[n]\rvert^2$
periodic CT power	$P = \frac{1}{T}\int_{\text{one period}} \lvert x(t)\rvert^2\,dt$
periodic DT power	$P = \frac{1}{N}\sum_{\text{one period}} \lvert x[n]\rvert^2$
sinusoid power	$P = \frac{A^2}{2}$
constant c power	$P = c^2$
flat piece, level A, length D	$A^2 D$
ramp $0 \to A$, length D	$\frac{A^2 D}{3}$
line $A \to B$, length D	$\frac{D}{3}\left(A^2 + AB + B^2\right)$
DT stem of height h	h^2

6. Transformations

shift	$x(t-t_0)$: right (delay) if $t_0 > 0$
reversal	$x(-t)$: mirror about $t = 0$
scaling	$x(at)$: compress if $\lvert a\rvert > 1$
point map for $x(at+b)$	$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$
undo $y(t) = x(at+b)$	$x(t) = y\left(\frac{t-b}{a}\right)$

7. Periodic signals

CT test	$x(t+T) = x(t)$
DT test	$x[n+N] = x[n]$, N an integer
DT sinusoid period	$\frac{\hat{\omega}}{2\pi} = \frac{m}{N}$ in lowest terms
sum of two periodic signals	lcm of the two periods

8. Even and odd

even test	$x(-t) = x(t)$
odd test	$x(-t) = -x(t)$
even part	$\mathcal{E}v\{x\} = \frac{x(t) + x(-t)}{2}$
odd part	$\mathcal{O}d\{x\} = \frac{x(t) - x(-t)}{2}$

9. Complex exponentials

CT form	$x(t) = Ce^{at}, \quad a = \sigma + j\omega$
real part	$\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi), \quad \phi = \angle C$
DT form	$x[n] = C\alpha^n$

10. Button, switch, ramp

DT switch → button	$\delta[n] = u[n] - u[n-1]$
DT button → switch	$u[n] = \sum_{k=-\infty}^{n} \delta[k]$
CT switch → button	$\delta(t) = \frac{du(t)}{dt}$
CT button → switch	$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$
area of the CT button	$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$
sifting (the flash)	$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$
DT sifting	$\sum_n x[n]\,\delta[n-n_0] = x[n_0]$
CT pulse from switches	$u(t-t_1) - u(t-t_2)$: on for $t_1 \le t < t_2$
DT pulse from switches	$u[n-a] - u[n-b]$: on for $a \le n \le b-1$
any DT signal as buttons	$x[n] = \sum_k x[k]\,\delta[n-k]$
ramp	$r(t) = t\,u(t), \quad r[n] = n\,u[n]$
ramp and switch, CT	$\frac{dr}{dt} = u(t)$
ramp and switch, DT	$r[n] - r[n-1] = u[n-1]$
graph from ramps	corner: $+(\Delta\text{slope})\,r(t-t_0)$

11–12. DT systems

unit delay D	$x[n] \to x[n-1]$
first difference	$y[n] = x[n] - x[n-1]$
3-point moving average	$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$
accumulator	$y[n] = y[n-1] + x[n]$
bounded input	$\lvert x[n]\rvert \le B < \infty$
stable if	$\lvert y[n]\rvert \le$ a finite number

13. Convolution

impulse response	$h[n] =$ output for $x = \delta$
convolution sum	$y[n] = \sum_k x[k]\,h[n-k]$
swap roles	$x * h = h * x$
button, delayed button	$x * \delta = x, \quad x * \delta[n-n_0] = x[n-n_0]$
output range, length	$[a+c, b+d], \quad L_x + L_h - 1$
sum check	$\sum y = \sum x \cdot \sum h$
causal, stable from h	$h[n<0] = 0, \quad \sum\lvert h\rvert < \infty$
series, parallel	$h_1 * h_2, \quad h_1 + h_2$
two exponentials	$\frac{a^{n+1} - b^{n+1}}{a-b}\,u[n]$

system property tests

Property	Ask yourself	Yes / no example
memoryless	Does y now use only x now?	yes x^2 no $x[n-1]$
causal	Does y avoid peeking at the future?	yes $x[n-1]$ no $x[n+1]$
stable	Does a capped input give a capped output?	yes x^2 no $\sum_{k\le n} x[k]$
invertible	Can I undo it and get x back?	yes $2x$ no x^2
time invariant	Does it ignore the clock? Delay in = delay out	yes $x[n-1]$ no $n\,x[n]$
linear	Scale in → scale out, add in → add out?	yes $n\,x[n]$ no $2x+3$

which tool? — given → find

Given → find	Do this	
$x(t)$, $T_s \to x[n]$	put $t = nT_s$	(1)
$A\cos(\omega t + \phi) \to f, T, t_0$	read, then divide	(2)
$f, f_s \to \hat{\omega}$	$\hat{\omega} = 2\pi f / f_s$	(3)
dies out or finite length	energy E_∞	(4)
periodic or constant	power P_∞	(4)
a graph → energy	square each piece, add	(5)
$x(t) \to x(at+b)$	map each breakpoint	(6)
$y = x(at+b) \to x$	$x(t) = y((t-b)/a)$	(6)
two graphs → $y = Ax(at+b)$	match two points, solve	(6)
signal → period	$2\pi/\omega$, or reduce $\hat{\omega}/2\pi$	(7)
signal → even, odd, neither	compare $x(-t)$ with $x(t)$	(8)
Ce^{at} → real signal	Euler, then the real part	(9)
integral with $\delta(t-t_0)$	the flash: three questions	(10)
a pulse graph → formula	a switch on, a switch off	(10)
DT stems → buttons	one $x[k]\,\delta[n-k]$ per stem	(10)
graph with corners → formula	a ramp per corner	(10)
blocks → one equation	substitute block by block	(11)
equation ↔ block diagram	one D per shift	(11)
system → properties	ask each question	(12)
system → stable?	cap the input with B	(12)
equation → h	read coefficients, or run $x = \delta$	(13)
$h \to$ causal, stable	$h[n<0] = 0, \quad \sum\lvert h\rvert$	(13)
stems x, $h \to y$	table: one row per stem	(13)
pulse and formula → y	flip and slide, by regions	(13)
two formulas → y	sum $k = 0..n$, geometric	(13)

The number in parentheses is the concept section.

Signals and Systems — Concepts

How to read a worked example. Every example is a puzzle: colored pieces go into the slots of the same color. One example works one signal or one system.

Step	What it does
1. PIECES	Say the problem **in words**. Read each value off the formula or graph and give it a color. Prep (sine to cosine, sorting terms) is named in this title.
2. TEMPLATE	The formula, with an empty slot [like this] for each piece, in the piece's color.
3. PLACE	Put each piece in its slot. Then do the arithmetic. A gray *(note)* says why each move is allowed.
middle steps	One move each, named by the move: SOLVE, LABEL, COMPARE.
last: CHECK	Test the answer, and say what the test proves.

Colors. Sinusoid colors stay the same everywhere: amplitude, frequency, phase or shift, sampling rate. In other examples, step 1 sets the colors.

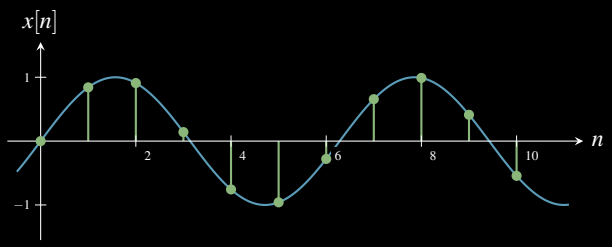
1. signals and sampling

THE IDEA
A **signal** is anything that changes over time and carries information. **Sampling** is taking snapshots: one photo every T_s seconds, like a strobe light.

- **Continuous time** (CT): $x(t)$, round brackets, any real t. The movie.
- **Discrete time** (DT): $x[n]$, square brackets, n an integer only. The stack of photos.

Trap: $x[2.5]$ does not exist. Between samples a DT signal is not zero. It is undefined.

Movie to photos: photo n is taken at $t = nT_s$, so $x[n] = x(nT_s)$. $T_s =$ **sampling period** (s between photos), $f_s = \frac{1}{T_s} =$ **sampling rate** (photos per s).



The curve is the movie (CT). The stems are the photos (DT).

Worked: sample $x(t) = t^2$ with $T_s = 0.5$ s

1. PIECES
In words: take one photo of the movie every half second.
the movie $x(t) = t^2$ the photo spacing $T_s = 0.5$

↓

2. TEMPLATE
$x[n] = [x(\cdot)]$ evaluated at $t = n \cdot [T_s]$

↓

3. PLACE
$x[n] = (n \cdot 0.5)^2 = 0.25n^2$ *(write $n \cdot 0.5$ wherever the movie says t)*

↓

4. LIST THE PHOTOS
$x[0] = 0,\quad x[1] = 0.25,\quad x[2] = 1,\quad x[3] = 2.25$

↓

5. CHECK
photo $n = 2$ is taken at $t = 1$ s, and $x(1) = 1^2 = 1$ ✓

Worked: (reverse) $x[n] = 0.25n^2$ came from $T_s = 0.5$ s. Find $x(t)$

1. PIECES
In words: undo the spacing. Photo n was taken at $t = nT_s$, so $n = t/T_s$.
the photos $x[n] = 0.25n^2$ the spacing $T_s = 0.5$

↓

2. TEMPLATE
$x(t) = [x[n]]$ with $n = \dfrac{t}{[T_s]}$

↓

3. PLACE
$n = \dfrac{t}{0.5} = 2t \quad \Rightarrow \quad x(t) = 0.25(2t)^2 = t^2$ *($(2t)^2 = 4t^2$)*

↓

4. CHECK
$t = 1$ is $n = 2$: $0.25(2)^2 = 1 = 1^2$ ✓

2. sinusoids

THE IDEA
A wheel spins at a steady speed with a dot on its rim. The dot's height over time is a sinusoid. Wheel size: amplitude. Spin rate: frequency. Where the dot starts: phase.

$$x(t) = [A]\cos\big([\omega]t + [\phi]\big)$$

- A: **amplitude**, the peak. It swings between $-A$ and A.
- ω: **angular frequency**, rad/s. $f = \frac{\omega}{2\pi}$ Hz (turns per s), $T = \frac{1}{f} = \frac{2\pi}{\omega}$ s per turn.
- ϕ: **phase**, rad. A head start or late start. As a **time shift**: $t_0 = -\frac{\phi}{\omega}$, $A\cos\big(\omega(t - t_0)\big)$.

Negative $\phi \Rightarrow t_0 > 0$: **delayed** (moves right).

Trap: Get the exact form $A\cos(\omega t + \phi)$ before reading pieces. Sine: $\sin\theta = \cos(\theta - \frac{\pi}{2})$. Negative A: $-A\cos\theta = A\cos(\theta + \pi)$.

Which way? — between the sinusoid numbers

Given → find	Do this
$\omega \to f$	divide by 2π: $f = \dfrac{\omega}{2\pi}$
$f \to \omega$	multiply by 2π: $\omega = 2\pi f$
$f \to T$	reciprocal: $T = \dfrac{1}{f}$
$T \to \omega$	$\omega = \dfrac{2\pi}{T}$
$\phi \to t_0$	$t_0 = -\dfrac{\phi}{\omega}$
$t_0 \to \phi$	$\phi = -\omega t_0$

Worked: read the parts of $x(t) = 3\cos\left(4\pi t - \frac{\pi}{3}\right)$

1. PIECES (MATCH THE FORM)
In words: a wheel of size 3, turning at 4π rad/s, starting $\frac{\pi}{3}$ late.
$A = 3 \qquad \omega = 4\pi \qquad \phi = -\frac{\pi}{3}$

↓

2. TEMPLATES
$f = \dfrac{[\omega]}{2\pi} \qquad T = \dfrac{1}{f} \qquad t_0 = -\dfrac{[\phi]}{[\omega]}$

↓

3. PLACE: FREQUENCY AND PERIOD
$f = \dfrac{4\pi}{2\pi} = 2$ Hz *(one turn is 2π rad)* $\qquad T = \dfrac{1}{2} = 0.5$ s

↓

4. PLACE: TIME SHIFT
$t_0 = -\dfrac{-\pi/3}{4\pi} = \frac{1}{12}$ s *(negative phase: delayed, moves right)*

↓

5. CHECK
rebuild it: $3\cos\left(4\pi(t - \frac{1}{12})\right)$, and $4\pi \cdot \frac{1}{12} = \frac{\pi}{3}$ ✓

Worked: (sine first) read the parts of $x(t) = 4\sin(10t)$

1. PIECES (SINE TO COSINE FIRST)
In words: a sine is a cosine a quarter turn late.
$4\sin(10t) = 4\cos\left(10t - \frac{\pi}{2}\right)$ $\quad A = 4, \quad \omega = 10, \quad \phi = -\frac{\pi}{2}$

↓

2. TEMPLATES
$f = \frac{\boxed{\omega}}{2\pi}$ $\quad T = \frac{2\pi}{\boxed{\omega}}$

↓

3. PLACE
$f = \frac{10}{2\pi} = 1.59$ Hz $\quad T = \frac{2\pi}{10} = 0.628$ s

↓

4. CHECK
$fT = 1.59 \times 0.628 = 1$ ✓ $\quad t = 0$: $4\sin 0 = 0 = 4\cos(-\frac{\pi}{2})$ ✓

Worked: (reverse) build $x(t)$: $A = 2$, $T = 0.4$ s, first peak at $t = 0.1$ s

1. PIECES
In words: a wheel of size 2, one turn every 0.4 s, at the top at $t = 0.1$.
$A = 2$ $\quad \omega = \frac{2\pi}{0.4} = 5\pi$ $\quad t_0 = 0.1 \to \phi = -\omega t_0 = -\frac{\pi}{2}$ *(the first peak is the shift)*

↓

2. TEMPLATE
$x(t) = \boxed{A}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$

↓

3. PLACE
$x(t) = 2\cos\left(5\pi t - \frac{\pi}{2}\right)$

↓

4. CHECK
$t = 0.1$: angle $\frac{\pi}{2} - \frac{\pi}{2} = 0$, so $x = 2\cos 0 = 2$, the peak ✓

3. DT sinusoids and DT frequency

THE IDEA
Photograph the spinning wheel with a strobe. In the photos you only see *how far the wheel turned between photos*. That angle per photo is the **DT frequency** $\hat{\omega}$, in rad/sample.

$$x[n] = A\cos(\hat{\omega}n + \phi), \qquad \hat{\omega} = \boxed{\omega}T_s = \frac{2\pi\boxed{f}}{\boxed{f_s}}$$

Aliasing is the wagon-wheel effect. A turn of $\frac{\pi}{4}$ per photo looks the same as $\frac{\pi}{4} + 2\pi$. So $\hat{\omega}$ and $\hat{\omega} + 2\pi$ give the same photos, and f and $f + f_s$ give the same samples. A cosine is even, so $-\hat{\omega}$ also gives the same photos: $f_s - f$ aliases to f too.

Which way? — CT frequency or DT frequency

Given → find	Do this
$f, f_s \to \hat{\omega}$	$\hat{\omega} = \frac{2\pi f}{f_s}$
$\omega, T_s \to \hat{\omega}$	$\hat{\omega} = \omega T_s$
$\hat{\omega}, f_s \to f$	$f = \frac{\hat{\omega} f_s}{2\pi}$ (the lowest answer)

Trap: DT to CT has more than one answer. Any alias $\pm f + k f_s$ fits the same photos.

Worked: sample $x(t) = \cos(200\pi t)$ at $f_s = 800$ Hz

1. PIECES
In words: the wheel turns 100 times a second, and the strobe flashes 800 times a second.
$\omega = 200\pi$ ($f = 100$ Hz) $\quad f_s = 800$

↓

2. TEMPLATE
$\hat{\omega} = \frac{\boxed{\omega}}{\boxed{f_s}}$ *(same as ωT_s)*

↓

3. PLACE
$\hat{\omega} = \frac{200\pi}{800} = \frac{\pi}{4}$ rad/sample $\Rightarrow$ $x[n] = \cos\left(\frac{\pi}{4}n\right)$

↓

4. CHECK
$\frac{\pi/4}{2\pi} = \frac{1}{8}$ turn per photo, so 8 photos per turn, and $\frac{800}{100} = 8$ ✓

Worked: (reverse) $\hat{\omega} = \frac{\pi}{4}$ at $f_s = 800$ Hz. Find f

1. PIECES
In words: the wheel turns $\frac{\pi}{4}$ per photo at 800 photos a second. How fast does it spin?
$\hat{\omega} = \frac{\pi}{4}$ $\quad f_s = 800$

↓

2. TEMPLATE
$f = \frac{\boxed{\hat{\omega}} \cdot \boxed{f_s}}{2\pi}$

↓

3. PLACE
$f = \frac{(\pi/4) \cdot 800}{2\pi} = 100$ Hz *(the lowest answer)*

↓

4. CHECK
$\frac{2\pi \cdot 100}{800} = \frac{\pi}{4}$ ✓

Worked: (alias) at $f_s = 800$ Hz, do 900 Hz and 700 Hz look like 100 Hz?

1. PIECES
In words: find each wheel's turn per photo, then remove whole turns.
$f_s = 800$ $\quad f = 900, \quad f = 700$

↓

2. TEMPLATE
$\hat{\omega} = \frac{2\pi\boxed{f}}{\boxed{f_s}}$, then subtract 2π until it lands in $(-\pi, \pi]$

↓

3. PLACE: 900 HZ
$\hat{\omega} = \frac{2\pi \cdot 900}{800} = \frac{9\pi}{4}, \quad -2\pi \to \frac{\pi}{4} \quad \Rightarrow$ same photos *($900 = 100 + f_s$)*

↓

4. PLACE: 700 HZ
$\hat{\omega} = \frac{7\pi}{4}, \quad -2\pi \to -\frac{\pi}{4} \quad \Rightarrow$ same photos *(cos is even, and $700 = f_s - 100$)*

↓

5. CHECK
$n = 1$: $\cos\frac{\pi}{4} = \cos\frac{9\pi}{4} = \cos\frac{7\pi}{4} = 0.707$ ✓

4. energy and power

THE IDEA
Think of $|x|^2$ as **money spent per day**, forever. **Energy** is your total spending. **Power** is your average per day. Spend \$100 once: total finite, average 0. Spend \$5 every day: total ∞, average \$5.

Why square? Picture x as a voltage on a $1\,\Omega$ resistor: $P = V^2/R = x^2$. Squaring also makes negatives count: $(-0.5)^2 = 0.25$. Without it, $\cos t$ would average to 0.
Total energy: add $|x|^2$ over all time, no dividing.

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2\,dt \qquad E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average power: total inside the window $[-T, T]$, divide by its length $2T$,

grow the window.

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2\,dt \qquad P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} |x[n]|^2$$

Periodic shortcut: one period tells the story, $P = \frac{1}{T}\int_{\text{period}} |x|^2\,dt$.

Puzzle template for energy: $E = \int_{\text{on at}}^{\text{off at}} |x|^2\,dt$ (DT: $\sum$). Where $x = 0$ it adds nothing, so the limits shrink to where the signal is on. A switch u tells you the on time.

Classify by shape first

The signal	Result
dies out, or finite length	**energy signal**: E_∞ finite, $P_\infty = 0$
periodic, or constant	**power signal**: P_∞ finite, $E_\infty = \infty$
grows without limit ($x = t$)	**neither**: both ∞

Energy signal: a fixed total E over a growing window, $\frac{E}{2T} \to 0$. Constant: energy grows as fast as the window, so T cancels: $\frac{c^2 \cdot 2T}{2T} = c^2$. A final E_∞ or P_∞ never contains T or N.

Traps from practice

Trap: **Square first**, then add. $E_\infty = 0$ only if $x = 0$ everywhere.
Trap: "16 forever" is the **power**. The energy is ∞.
Trap: t disappears when you integrate. T is what goes to ∞. $\frac{c}{T} \to 0$, not c. Write dt, not Δt.
Trap: **Write the label**: energy signal, power signal, or neither.

Worked: pulse: $x(t) = 2$ for $0 \le t \le 3$, else 0

1. PIECES
In words: height 2, on for 3 s, then off for good.
on at 0, off at 3, $|x|^2 = 4$ while on

↓

2. TEMPLATE
$E = \int_{\text{on}}^{\text{off}} |x|^2\,dt$

↓

3. PLACE
$E = \int_0^3 4\,dt = 4(3) = 12$ *(where $x = 0$ it adds nothing)*

↓

4. LABEL
$P = \lim_{T\to\infty} \frac{12}{2T} = 0$ *(a fixed total over a growing window)* $\Rightarrow$ energy signal

↓

5. CHECK (SHORTCUT A^2D)
$2^2 \cdot 3 = 12$ ✓

Worked: decaying exponential: $x(t) = 3e^{-2t}u(t)$

1. PIECES
In words: switched on at 0, never off, shrinking toward 0.
on at 0, off never, $|x|^2 = 9e^{-4t}$ *(squaring doubles the exponent)*

↓

2. TEMPLATE
$E = \int_{\text{on}}^{\text{off}} |x|^2\,dt$

↓

3. PLACE
$E = \int_0^\infty 9e^{-4t}\,dt = 9 \cdot \frac{1}{4} = 2.25$ $(\int_0^\infty e^{-at}\,dt = \frac{1}{a})$

↓

4. LABEL
finite energy $\Rightarrow$ energy signal

↓

5. CHECK
antiderivative $-\frac{9}{4}e^{-4t}$: $0 - \left(-\frac{9}{4}\right) = 2.25$ ✓

Worked: DT geometric: $x[n] = \left(\frac{1}{2}\right)^n u[n]$

1. PIECES
In words: starts at 1 at $n = 0$, and each stem is half the last.
on at 0, off never, $|x|^2 = \left(\frac{1}{2}\right)^{2n} = \left(\frac{1}{4}\right)^n$

↓

2. TEMPLATE
$E = \sum_{\text{on}}^{\text{off}} |x|^2$

↓

3. PLACE
$E = \sum_0^\infty \left(\frac{1}{4}\right)^n = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$ *(geometric sum, $r = \frac{1}{4}$)*

↓

4. LABEL
finite energy $\Rightarrow$ energy signal

↓

5. CHECK
$1 + 0.25 + 0.0625 + 0.0156 + \ldots \approx 1.33$ ✓

Worked: sinusoid: $x(t) = 4\cos(3t)$

1. PIECES
In words: it repeats forever, so find the power.
$A = 4$ $\qquad |x|^2 = 16\cos^2(3t) = 8 + 8\cos(6t)$ $(\cos^2\theta = \frac{1+\cos 2\theta}{2})$

↓

2. TEMPLATE
$P =$ average of $|x|^2$ over one period

↓

3. PLACE
average of 8 is 8, average of $8\cos(6t)$ is 0 *(a cosine is up as much as down)* $\Rightarrow P = 8$

↓

4. LABEL
it never dies out: $E = \infty$ $\Rightarrow$ power signal

↓

5. CHECK (SHORTCUT $\frac{A^2}{2}$)
$\frac{4^2}{2} = 8$ ✓

Worked: constant: $x(t) = 3$

1. PIECES
In words: always 3, never dies out, so find the power.
$|x|^2 = 9$ for all t

↓

2. TEMPLATE
$P = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x|^2\,dt$

↓

3. PLACE: WINDOW TOTAL
$\int_{-T}^{T} 9\,dt = 9T - (-9T) = 18T$

↓

4. DIVIDE BY THE WINDOW
$P = \frac{18T}{2T} = 9$ *(T cancels, so the limit is 9)*

↓

5. LABEL
$E = \lim 18T = \infty$ $\Rightarrow$ power signal

↓

6. CHECK
a constant c has $P = c^2$: $3^2 = 9$ ✓

Quick example	Work → answer
DT $x[n] = (-1)^n$	$\lvert x\rvert^2 = 1$, $2N+1$ terms: $P = \frac{2N+1}{2N+1} = 1$, power signal
$x(t) = t$	grows without limit: neither

5. energy from a graph

THE IDEA
Do not set up one big integral. Chop the graph into pieces, find each piece's energy from the shortcut table, and add. In DT there are no widths: each stem's energy is its height squared.

Shortcuts (piece of length D) — these are the templates:

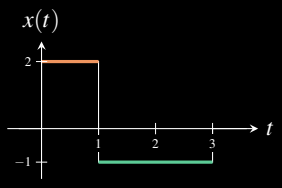
Piece of the graph	Energy of that piece
flat at level A	A^2D
ramp from 0 to A (or A to 0)	$\frac{A^2D}{3}$
line from A to B	$\frac{D}{3}(A^2+AB+B^2)$
DT stem of height h	h^2

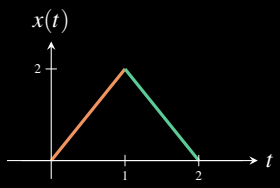
Line through (t_1,x_1), slope m: $x = x_1 + m(t-t_1)$. Use a shortcut to check an integral, and the reverse.

Trap: Negative parts count positive: $(-2)^2 = 4$.

Trap: A squared ramp is a curve, not a triangle: $\frac{A^2D}{3}$, not $\frac{A^2D}{2}$.

Trap: If the graph repeats forever, $E_\infty = \infty$. Find the power over one period.





Worked: two flat pieces (left graph)

1. PIECES
In words: height 2 for 1 s, then height -1 for 2 s.
flat at 2, $D=1$ flat at -1, $D=2$

↓

2. TEMPLATE
$E = \boxed{A^2D} + \boxed{A^2D}$

↓

3. PLACE
$E = 2^2(1) + (-1)^2(2) = 4+2 = 6$ *($(-1)^2=1$: negatives count)*

↓

4. CHECK BY INTEGRAL
$\int_0^1 4\,dt + \int_1^3 1\,dt = 4+2 = 6$ ✓

Worked: a triangle (right graph)

1. PIECES
In words: a straight rise from 0 to 2, then a straight fall back to 0.
ramp $0\to 2$, $D=1$ ramp $2\to 0$, $D=1$

↓

2. TEMPLATE
$E = \frac{\boxed{A}^2\boxed{D}}{3} + \frac{\boxed{A}^2\boxed{D}}{3}$

↓

3. PLACE
$E = \frac{2^2(1)}{3} + \frac{2^2(1)}{3} = \frac{8}{3}$ *(a squared ramp is a curve: ÷3, not ÷2)*

↓

4. CHECK BY INTEGRAL
rise $x = 2t$: $\int_0^1 4t^2\,dt = \frac{4}{3}$. The fall is its mirror, $\frac{4}{3}$ ✓

Worked: a line from 1 to 3 over $[0,2]$

1. PIECES
In words: one straight piece that starts at 1 and ends at 3.
$A=1$ $B=3$ $D=2$

↓

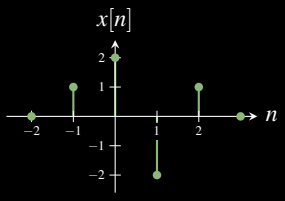
2. TEMPLATE
$E = \frac{\boxed{D}}{3}(\boxed{A}^2 + \boxed{A}\boxed{B} + \boxed{B}^2)$

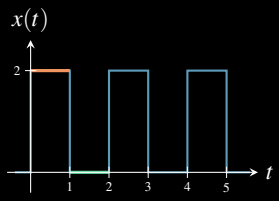
↓

3. PLACE
$E = \frac{2}{3}(1+1\cdot 3+9) = \frac{2}{3}(13) = \frac{26}{3}$

↓

4. CHECK BY INTEGRAL
$x = 1+t$: $\int_0^2 (1+2t+t^2)\,dt = 2+4+\frac{8}{3} = \frac{26}{3}$ ✓





Worked: a DT stem plot (left graph)

1. PIECES (EVERY STEM)
In words: no widths in DT, so square each stem height and add.
heights $1, 2, -2, 1$ at $n = -1, 0, 1, 2$

↓

2. TEMPLATE
$E = \sum \boxed{h}^2$

↓

3. PLACE
$E = 1^2 + 2^2 + (-2)^2 + 1^2 = 1+4+4+1 = 10$

↓

4. LABEL
$P = \lim_{N\to\infty} \frac{10}{2N+1} = 0 \quad \Rightarrow$ energy signal

↓

5. CHECK
4 stems, each counted once, the negative squared to $+4$ ✓

Worked: a periodic graph (right graph): power, not energy

1. PIECES (ONE PERIOD, $T=2$)
In words: it repeats every 2 s, so find the power over one period.
flat at 2, $D=1$ flat at 0, $D=1$

↓

2. TEMPLATE
$P = \frac{\boxed{A^2D} + \boxed{A^2D}}{T}$

↓

3. PLACE
$P = \frac{4+0}{2} = 2$

↓

4. LABEL
it repeats forever: $E = \infty \quad \Rightarrow$ power signal

↓

5. CHECK
on half the time at $x^2 = 4$: average $\frac{4}{2} = 2$ ✓

6. transformations of the time variable

THE IDEA
You are not changing the shape, only *when* things happen. Ask one question of every corner of the graph: **where did this point move to?** It keeps its height and lands at a new time.

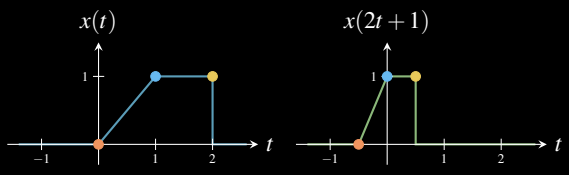
- **Shift** $x(t-t_0)$: $t_0 > 0$ moves right (a **delay**). $t_0 < 0$ moves left (an **advance**).
- **Reversal** $x(-t)$: play the tape backwards, mirror about $t = 0$.

- **Scaling** $x(at)$: fast-forward (compress) if $|a|>1$. Slow motion (stretch) if $|a|<1$.

Point map for $x(at+b)$ — the template, one step, no order of operations:

$$t_{\text{new}} = \frac{\boxed{t_{\text{old}}} - b}{a}$$

Trap: Scale-then-shift is wrong: $x(2t)$ shifted left 1 is $x(2(t+1)) = x(2t+2)$, not $x(2t+1)$.
Trap: The sign inside is opposite to the direction: $x(t-3)$ moves *right*.
Trap: DT scaling loses photos: $x[2n]$ keeps only the even-index samples.



Worked: sketch $x(2t+1)$ from $x(t)$ (the dots move)

1. PIECES
In words: every corner keeps its height and moves to a new time.
corners $t=0$ (height 0), $t=1$ (height 1), $t=2$ (height 1, then drops) $a=2$, $b=1$

↓

2. TEMPLATE
$$t_{\text{new}} = \frac{\boxed{t_{\text{old}}} - 1}{2}$$

↓

3. PLACE EACH DOT
$\frac{0-1}{2} = -\frac{1}{2}$ $\frac{1-1}{2} = 0$ $\frac{2-1}{2} = \frac{1}{2}$ *(heights unchanged)*

↓

4. DRAW
connect the dots in the same order: a ramp on $[-\frac{1}{2}, 0]$, flat at 1 until $\frac{1}{2}$, then a drop to 0

↓

5. CHECK
$t=0$: $x(2\cdot 0+1) = x(1) = 1$, the blue dot ✓

Worked: (reversal) sketch $x(-t+2)$ from the same $x(t)$

1. PIECES
In words: play the tape backwards, then slide it. Each corner still keeps its height.
corners 0, 1, 2 $a=-1$, $b=2$

↓

2. TEMPLATE
$$t_{\text{new}} = \frac{\boxed{t_{\text{old}}} - 2}{-1} = 2 - \boxed{t_{\text{old}}}$$

↓

3. PLACE EACH DOT
$0 \to 2$ $1 \to 1$ $2 \to 0$

↓

4. DRAW
flat at 1 on $[0,1]$, then a ramp $1 \to 0$ on $[1,2]$: the mirror image

↓

5. CHECK
$t=1.5$: $x(-1.5+2) = x(0.5) = 0.5$, on the ramp ✓

Worked: (DT compress) $x[0]=1$, $x[1]=2$, $x[2]=3$, else 0. Find $x[2n]$

1. PIECES
In words: keep every other photo.
samples at 0, 1, 2 $a=2$, $b=0$

↓

2. TEMPLATE
$n_{\text{new}} = \frac{\boxed{n_{\text{old}}}}{2}$, and it must be an integer

↓

3. PLACE EACH SAMPLE
$0 \to 0$ $2 \to 1$ $1 \to \frac{1}{2}$: not an integer, lost

↓

4. RESULT
$x[2n]$: 1 at $n=0$, 3 at $n=1$, else 0

↓

5. CHECK
$n=1$: $x[2\cdot 1] = x[2] = 3$ ✓ *(the 2 is gone for good: $x[2n]$ cannot be undone)*

Quick example	Work → answer
delay $x[n-2]$, same x	each sample $+2$: $0,1,2 \to 2,3,4$
reverse $x[-n]$, same x	each sample flips: $0,1,2 \to 0,-1,-2$

Which way? — undo a transformation

Given → find	Do this
$x(t) \to x(at+b)$	map each breakpoint: $t_{\text{new}} = \frac{t_{\text{old}}-b}{a}$
$y(t) = x(at+b) \to x(t)$	$x(t) = y\left(\frac{t-b}{a}\right)$

Trap: $x[2n]$ cannot be undone. The odd-index photos were thrown away.

Worked: (reverse) $y(t) = x(2t+1)$. Find $x(t)$

1. PIECES
In words: undo the map. Solve $2t+1=s$ for t, then rename s to t.
$a=2$ $b=1$

↓

2. TEMPLATE
$$x(t) = y\left(\frac{t-\boxed{b}}{\boxed{a}}\right)$$

↓

3. PLACE
$$x(t) = y\left(\frac{t-1}{2}\right)$$

↓

4. CHECK
$y\left(\frac{t-1}{2}\right) = x\left(2\cdot\frac{t-1}{2}+1\right) = x(t)$ ✓

Which way? — find $y = Ax(at+b)$ from two graphs

$$y(t) = \boxed{A}\,x\left(\boxed{a}\,t + \boxed{b}\right)$$

1. $A = \frac{\text{height of } y}{\text{height of } x}$. Upside down means $A<0$.
2. Mirrored? Then $a<0$.
3. Two matching points: x at t_1, t_2 moved to τ_1, τ_2. Solve $a\tau_1 + b = t_1$, $a\tau_2 + b = t_2$ *(y at τ is x at $a\tau+b$).*
4. Check a third point.

Worked: x: ramp $0 \to 1$ on $[0,4]$. y: same ramp, height 3, on $[-1,1]$

1. PIECES
In words: y is x made 3 times taller and squeezed onto $[-1,1]$, not flipped.
$A = \frac{3}{1} = 3$ start $t=0$ moved to $\tau=-1$, end $t=4$ moved to $\tau=1$

↓

2. TEMPLATE
$\boxed{a}\tau + \boxed{b} = t$ for each matching pair

↓

3. PLACE, SOLVE
start: $a(-1)+b=0$ end: $a(1)+b=4$ add: $b=2$, then $a=2$

↓

4. RESULT
$y(t) = 3x(2t+2)$

↓

5. CHECK (A THIRD POINT)
$t=0$: $3x(2) = 3(0.5) = 1.5$, half of 3 ✓

Worked: (mirrored) same x, and y falls $3 \to 0$ on $[-1,1]$

1. PIECES
In words: same as before, but flipped left to right.
$A = 3$ start $t = 0$ moved to $\tau = 1$, end $t = 4$ moved to $\tau = -1$

↓

2. TEMPLATE
$\boxed{a}\tau + \boxed{b} = t$ for each matching pair

↓

3. PLACE, SOLVE
start: $a(1) + b = 0$ end: $a(-1) + b = 4$ $a = -2, b = 2$

↓

4. RESULT
$y = 3x(-2t + 2)$

↓

5. CHECK
$t = -1$: $3x(4) = 3$, the top ✓ $a < 0$, as a mirror needs ✓

Worked: (DT) $x[0..2] = 1, 2, 3$, and $y[-1] = 3$, $y[0] = 2$, $y[1] = 1$

1. PIECES
In words: same heights, read backwards.
$A = 1$ mirrored: $a = -1$ one match: $y[-1] = 3 = x[2]$

↓

2. TEMPLATE
$y[n] = x[\boxed{a}n + \boxed{b}]$

↓

3. PLACE, SOLVE
$x[-1(-1) + b] = x[2] \Rightarrow 1 + b = 2, \quad b = 1$

↓

4. RESULT
$y[n] = x[-n + 1]$

↓

5. CHECK
$n = 0$: $x[1] = 2$ ✓ $n = 1$: $x[0] = 1$ ✓

7. periodic signals

THE IDEA
A periodic signal is a song on repeat. Slide it by one period and nothing changes. The **fundamental period** is the shortest loop.

$$x(t) = x(t + T) \qquad x[n] = x[n + N],\ N \text{ an integer}$$

- **CT sinusoid**: always periodic, $T = \dfrac{2\pi}{\boxed{\omega}}$.
- **DT sinusoid**: periodic only if a whole number of photos holds a whole number of turns: $\dfrac{\boxed{\hat{\omega}}}{2\pi} = \dfrac{m}{\boxed{N}}$ in lowest terms.
- **Sum of two**: the loops realign after $\text{lcm}(T_1, T_2)$.

Trap: $\cos(n)$ is not periodic ($\frac{1}{2\pi}$ is irrational), though $\cos(t)$ is. Sampling can destroy periodicity.

Worked: period of $\cos\left(\frac{2\pi}{3}t\right) + \sin\left(\frac{\pi}{2}t\right)$

1. PIECES
In words: two songs on repeat. Find when both loops line up again.
$\omega_1 = \frac{2\pi}{3}$ $\omega_2 = \frac{\pi}{2}$

↓

2. TEMPLATE
$T_1 = \dfrac{2\pi}{\boxed{\omega_1}}, \quad T_2 = \dfrac{2\pi}{\boxed{\omega_2}}, \quad T = \text{lcm}(T_1, T_2)$

↓

3. PLACE
$T_1 = \dfrac{2\pi}{2\pi/3} = 3$ $T_2 = \dfrac{2\pi}{\pi/2} = 4$

↓

4. LCM
$\text{lcm}(3, 4) = 12$ s *(the first time both loops restart together)*

↓

5. CHECK
12 s holds 4 whole cycles of the first and 3 of the second ✓

Worked: DT period of $\cos\left(\frac{3\pi}{8}n\right)$

1. PIECES
In words: find the fewest photos that hold a whole number of turns.
$\hat{\omega} = \frac{3\pi}{8}$

↓

2. TEMPLATE
$\dfrac{\boxed{\hat{\omega}}}{2\pi} = \dfrac{m}{\boxed{N}}$, reduced

↓

3. PLACE
$\dfrac{3\pi/8}{2\pi} = \dfrac{3}{16}$, already reduced $\Rightarrow N = 16$

↓

4. CHECK
$\frac{3\pi}{8} \cdot 16 = 6\pi$: exactly 3 whole turns ✓

Quick example	Work → answer
$\cos\left(\frac{4\pi}{6}n\right)$	$\frac{4}{12} = \frac{1}{3} \Rightarrow N = 3$ not 12: reduce first
$\cos(n)$	$\frac{1}{2\pi}$ is not a fraction of integers: not periodic

8. even and odd signals

THE IDEA
Even: a mirror on the vertical axis. The left half reflects the right. **Odd**: spin the graph half a turn about the origin and it lands on itself.

- Even: $x(-t) = x(t)$. $\cos$, t^2, constants.
- Odd: $x(-t) = -x(t)$. $\sin$, t, t^3.

Test: replace t with $-t$, simplify, compare. **Split**:

$$\mathcal{E}v\{x\} = \frac{x(t) + x(-t)}{2} \qquad \mathcal{O}d\{x\} = \frac{x(t) - x(-t)}{2}$$

Shortcut for sums of terms: sort each term into a even or odd pile. The piles *are* the two parts. Products work like signs: even·even = odd·odd = even, and even·odd = odd.
Trap: An odd signal is 0 at the origin: $x(0) = -x(0)$.

Worked: is $x(t) = t^3 - 2t$ even, odd, or neither?

1. PIECES (SORT TERMS)
In words: put each term in the even pile or the odd pile.
t^3 odd $-2t$ odd no even terms

↓

2. TEMPLATE
$x(t) = \boxed{\text{even pile}} + \boxed{\text{odd pile}}$

↓

3. PLACE
$x = 0 + (t^3 - 2t) \Rightarrow$ odd *(only the odd pile has terms)*

↓

4. CHECK BY TEST
$x(-t) = -t^3 + 2t = -(t^3 - 2t) = -x(t)$ ✓

Worked: $x(t) = t^2 + t$: test, then split

1. PIECES (SORT TERMS)
In words: one term in each pile, so it is neither, and the piles are the parts.
t^2 even t odd

↓

2. TEMPLATE
$\mathcal{E}v = \boxed{\text{even pile}}, \qquad \mathcal{O}d = \boxed{\text{odd pile}}$

↓

3. PLACE
$\mathcal{E}v = t^2$ $\mathcal{O}d = t$ both piles non-empty $\Rightarrow$ neither

↓

4. CHECK BY FORMULA
$\dfrac{(t^2 + t) + (t^2 - t)}{2} = t^2$ ✓ $\dfrac{(t^2 + t) - (t^2 - t)}{2} = t$ ✓

Worked: DT split: $x[0]=1$, $x[1]=2$, $x[2]=3$, ***else*** 0

1. PIECES
In words: no formula, so build a table. Read $x[-n]$ from x at the opposite index.
$x[n]$ for $n=-2..2$: $0,0,1,2,3$ $\quad$ $x[-n]$: $3,2,1,0,0$

↓

2. TEMPLATE
$\frac{x[n]+x[-n]}{2}$ $\quad$ $\frac{x[n]-x[-n]}{2}$, one column per n

↓

3. PLACE

n	-2	-1	0	1	2
$x[n]$	0	0	1	2	3
$x[-n]$	3	2	1	0	0
even	1.5	1	1	1	1.5
odd	-1.5	-1	0	1	1.5

↓

4. CHECK EACH COLUMN
even + odd $=x[n]$: $n=2$: $1.5+1.5=3$ ✓ $n=-2$: 0 ✓ $\quad$ odd part is 0 at $n=0$ ✓

9. complex exponentials

THE IDEA
$e^{j\theta}$ is an arrow of length 1 pointing at angle θ. Let $\theta=\omega t$ and it spins. Its shadow on the horizontal axis is a cosine. Multiply by $e^{\sigma t}$ and it also grows or shrinks: a spiral.

Euler: $e^{j\theta}=\cos\theta+j\sin\theta$. $\quad \cos\theta=\dfrac{e^{j\theta}+e^{-j\theta}}{2}$, $\quad \sin\theta=\dfrac{e^{j\theta}-e^{-j\theta}}{2j}$

(two arrows spinning opposite ways).

CT form: $x(t)=Ce^{at}$, $\quad C=\boxed{|C|}e^{j\boxed{\phi}}$ (start length, angle), $\quad a=\boxed{\sigma}+j\boxed{\omega}$ (growth, spin). The template:

$$\mathrm{Re}\{Ce^{at}\}=\boxed{|C|}e^{\boxed{\sigma}t}\cos\left(\boxed{\omega}t+\boxed{\phi}\right)$$

Case	Real part looks like
$\omega=0$	no spin: **real exponential**. Grows if $\sigma>0$, decays if $\sigma<0$
$\sigma=0$	pure spin: **sinusoid**, $T=\frac{2\pi}{\lvert\omega\rvert}$
both	spiral: **damped sinusoid**, envelope $\lvert C\rvert e^{\sigma t}$

DT form: $x[n]=C\alpha^n$. Each step multiplies by α. $|\alpha|<1$ decays, $|\alpha|>1$ grows, $\alpha<0$ flips sign each step.

DT frequencies repeat every 2π: $e^{j(\hat\omega+2\pi)n}=e^{j\hat\omega n}$, because $e^{j2\pi n}=1$ for integer n.

CT $e^{j\omega t}$	DT $e^{j\hat\omega n}$
each ω a different signal	$\hat\omega$ and $\hat\omega+2\pi$ the same
always periodic	periodic only if $\frac{\hat\omega}{2\pi}$ is rational
larger $\omega\Rightarrow$ faster	fastest at $\hat\omega=\pi$ $(+1,-1,\ldots)$

Worked: real part of $x(t)=2e^{j\pi/3}e^{(-1+j4)t}$

1. PIECES
In words: an arrow of length 2, starting at $\frac{\pi}{3}$, spinning at 4 rad/s, shrinking like e^{-t}.
C: $|C|=2$, $\phi=\frac{\pi}{3}$ $\quad$ a: $\sigma=-1$, $\omega=4$

↓

2. TEMPLATE
$\mathrm{Re}\{x\}=\boxed{|C|}e^{\boxed{\sigma}t}\cos\left(\boxed{\omega}t+\boxed{\phi}\right)$

↓

3. PLACE
$\mathrm{Re}\{x\}=2e^{-1t}\cos\left(4t+\frac{\pi}{3}\right)$ *(a damped sinusoid with envelope $2e^{-t}$)*

↓

4. CHECK
$t=0$: $\mathrm{Re}\{2e^{j\pi/3}\}=2\cos\frac{\pi}{3}=1$, and the answer gives $2\cdot1\cdot\frac{1}{2}=1$ ✓

Worked: (reverse) write $5e^{-2t}\cos\left(3t-\frac{\pi}{4}\right)$ ***as*** $\mathrm{Re}\{Ce^{at}\}$

1. PIECES (MATCH THE TEMPLATE)
In words: read the length, growth, spin, and start angle off the real signal.
$|C|=5$ $\quad$ $\sigma=-2$ $\quad$ $\omega=3$ $\quad$ $\phi=-\frac{\pi}{4}$

↓

2. TEMPLATE
$C=\boxed{|C|}e^{j\boxed{\phi}}$ $\quad$ $a=\boxed{\sigma}+j\boxed{\omega}$

↓

3. PLACE
$C=5e^{j(-\pi/4)}=5e^{-j\pi/4}$ $\quad$ $a=-2+j3=-2+j3$

↓

4. CHECK
$\mathrm{Re}\{5e^{-j\pi/4}e^{(-2+j3)t}\}=5e^{-2t}\cos\left(3t-\frac{\pi}{4}\right)$ ✓

Worked: write $6\cos(5t)$ ***as two complex exponentials***

1. PIECES
In words: a cosine is two arrows spinning opposite ways, each half as long.
$A=6$ $\quad$ $\omega=5$

↓

2. TEMPLATE
$\boxed{A}\cos(\boxed{\omega}t)=\boxed{A}\cdot\dfrac{e^{j\boxed{\omega}t}+e^{-j\boxed{\omega}t}}{2}$

↓

3. PLACE
$6\cdot\dfrac{e^{j5t}+e^{-j5t}}{2}=3e^{j5t}+3e^{-j5t}$

↓

4. CHECK
$t=0$: $3+3=6=6\cos0$ ✓

Quick example	Work → answer
$\cos\left(\frac{9\pi}{4}n\right)$	remove a whole turn: $\frac{9\pi}{4}-2\pi=\frac{\pi}{4}\Rightarrow\cos\left(\frac{\pi}{4}n\right)$, $N=8$
$x[n]=(-0.5)^n$	$\alpha=-0.5$: shrinks and flips sign each step: $1,-0.5,0.25,\ldots$

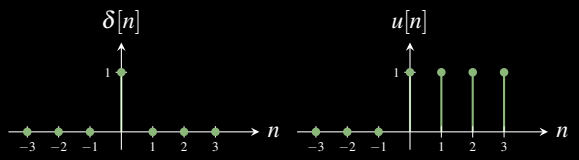
10. the button, the switch, and the ramp

THE IDEA

- δ is a **button**: press once at time 0, then release.
- u is a **switch**: flip it on at time 0 and leave it on.
- r is a **ramp**: the running total of the switch, climbing $0,1,2,3,\ldots$

Each is the running total of the one before. Everything else is built from these three.

$$\delta[n]=\begin{cases}1 & n=0\\ 0 & n\neq0\end{cases}\qquad u[n]=\begin{cases}1 & n\geq0\\ 0 & n<0\end{cases}$$



Trap: The switch does not keep climbing: $u[n]$ is $0,0,1,1,1,\ldots$ One stair up, then flat. The ramp is what climbs.

Size: the button is one stem of height 1, $E_\infty=1$ (energy signal). The switch stays on forever: $E_\infty=\infty$, finite power (power signal).

Running totals and where the button sits

Switch = running total of presses: $u[n]=\sum_{k=-\infty}^{n}\delta[k]$ *(no presses before 0, one press at 0, the total stays 1)*. **Back**: a switch on at 0 minus a switch on at 1 is on only at 0: $\delta[n]=u[n]-u[n-1]$.

Where is it? Set the inside to 0 and solve. $\delta[n-k]$ sits at $n=k$. $u[n-k]$ turns on at $n=k$.

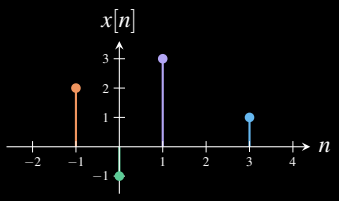
Inside	Stem at	Moves	Name
$n-3$	$n=3$	right	**delay**: later
$n+2$	$n=-2$	left	**advance**: earlier

Trap: The sign inside is opposite to the position, and $+2$ is an *advance*.

Building signals: switches on and off, one button per stem

Pulse: $u[n-\boxed{a}]-u[n-\boxed{b}]$ is on for $\boxed{a}\le n\le \boxed{b}-1$ ($b-a$ samples). CT: $u(t-t_1)-u(t-t_2)$ is on for $t_1\le t<t_2$.
Trap: The off-switch's own position is already off: $u[n]-u[n-4]$ at $n=4$ is $1-1=0$.
One button per stem: a stem of height h at position k is $\boxed{h}\,\delta[n-\boxed{k}]$. Add one term per stem:

$$x[n]=\sum_k x[k]\,\delta[n-k]$$



Worked: write the stems above with buttons

1. PIECES (HEIGHT AT POSITION)
In words: one button per stem, as tall as the stem, pressed at its position.
2 at -1 $\quad -1$ at 0 $\quad$ 3 at 1 $\quad$ 1 at 3 *(no stem at 2: no term)*

↓

2. TEMPLATE (ONE PER STEM)
$\boxed{h}\,\delta[n-\boxed{k}]+\cdots$

↓

3. PLACE
$x[n]=2\delta[n+1]-\delta[n]+3\delta[n-1]+\delta[n-3]$ *($n-(-1)=n+1$)*

↓

4. CHECK
$n=1$: only $\delta[n-1]$ is on, so $x[1]=3$ ✓ $\quad n=2$: all off, 0 ✓

Worked: (reverse) read the stems of $x[n]=4\delta[n-2]-2\delta[n+3]$

1. PIECES
In words: each button is one stem. Its number is the height. Where its inside is 0 is the position.
$4\delta[n-2]$ $\quad -2\delta[n+3]$

↓

2. TEMPLATE
coefficient = height $\quad$ inside $=0\Rightarrow$ position

↓

3. PLACE
$n-2=0$: $x[2]=4$ $\quad n+3=0$: $x[-3]=-2$ *(+3 is an advance: left)*

↓

4. CHECK
$n=2$: $4(1)-2(0)=4$ ✓

Worked: DT pulse $x[n]=u[n]-u[n-3]$

1. PIECES
In words: a switch on at 0, and a second switch that cancels it from 3 on.
on at $a=0$, $\quad$ off at $b=3$

↓

2. TEMPLATE
on for $\boxed{a}\le n\le \boxed{b}-1$

↓

3. PLACE
on for $0\le n\le 3-1=2 \;\Rightarrow\; x=\delta[n]+\delta[n-1]+\delta[n-2]$

↓

4. FIRST DIFFERENCE
a button at each switch: $x[n]-x[n-1]=\delta[n]-\delta[n-3]$

↓

5. CHECK
$n=3$: $u[3]-u[0]=1-1=0$, already off ✓

The CT button

A one-instant press does not exist in CT. Picture a rectangle of width ε, height $\frac{1}{\varepsilon}$: area 1 always. Shrink $\varepsilon\to 0$: infinitely tall, area still 1. A hammer tap: instant, but a definite amount of hit.

$$\int_{-\infty}^{\infty}\delta(t)\,dt=1 \qquad \text{(CT: area 1, DT: height 1)}$$

CT switch: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$. Back: $\delta(t)=\frac{du}{dt}$. A jump in a signal becomes a button in its derivative.
Trap: δ is a function of t, not a variable. $\int\delta\,dt$ is not δ^2. Use its one fact, area 1.

Sifting: the camera flash

Fire a flash at t_0 and see what value of g it captures. The template:

$$\int_{\boxed{\text{lo}}}^{\boxed{\text{hi}}}\boxed{g(t)}\,\delta\big(t-\boxed{t_0}\big)\,dt=\begin{cases}\boxed{g}(\boxed{t_0}) & \boxed{t_0}\text{ inside }[\boxed{\text{lo}},\boxed{\text{hi}}]\\ 0 & \text{outside}\end{cases}$$

Three questions: (1) Where is the flash? Set the inside of δ to 0. (2) In the window? If not, the answer is 0. (3) What does it capture? Plug t_0 into everything except the δ. The answer has no δ left.
Sampling form: $x(t)\,\delta(t-t_0)=x(t_0)\,\delta(t-t_0)$. DT is the same with n_0.

Worked: $\int_{-2}^{6}(3t+1)\,\delta(t-4)\,dt$

1. PIECES (THE THREE QUESTIONS)
In words: a flash at one instant captures $3t+1$ there, if the flash is inside the window.
flash at $t=4$ *($t-4=0$)* $\quad$ window $[-2,6]$ $\quad g=3t+1$

↓

2. TEMPLATE
$\boxed{\text{flash}}$ in $\boxed{\text{window}}$? $\quad$ yes: $\boxed{g}(\boxed{\text{flash}})$ $\quad$ no: 0

↓

3. PLACE
4 is in $[-2,6] \;\Rightarrow\; 3(4)+1=13$

↓

4. CHECK
no δ is left in the answer ✓

Worked: (outside the window) $\int_{1}^{3}5\,\delta(t)\,dt$

1. PIECES (THE THREE QUESTIONS)
In words: the flash fires at 0, but the window starts at 1.
flash at $t=0$ $\quad$ window $[1,3]$ $\quad g=5$

↓

2. TEMPLATE
$\boxed{\text{flash}}$ in $\boxed{\text{window}}$? $\quad$ yes: $\boxed{g}(\boxed{\text{flash}})$ $\quad$ no: 0

↓

3. PLACE
$0\notin[1,3] \;\Rightarrow\; 0$

↓

4. CHECK
over all t it would be 5. The window decides ✓

Quick example	Work → answer
$\int_{-\infty}^{\infty}(t^2+1)\,\delta(t-2)\,dt$	flash at 2, in: $2^2+1=5$
$\int_0^5 e^t\,\delta(t+1)\,dt$	flash at -1, outside $[0,5]$: 0
DT $\sum_n n^2\delta[n-3]$	flash at 3, all n: $3^2=9$

Which way? — switch or button

Given → find	Do this
$u\to\delta$ (CT)	differentiate: $\delta(t)=\frac{du}{dt}$
$\delta\to u$ (CT)	integrate: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$
$u\to\delta$ (DT)	first difference: $\delta[n]=u[n]-u[n-1]$
$\delta\to u$ (DT)	running sum: $u[n]=\sum_{k=-\infty}^{n}\delta[k]$

Trap: CT uses derivatives and integrals. DT uses differences and sums.

Worked: pulse of height 2 on $[1,4)$ with switches, then down and back

1. PIECES
In words: one switch turns 2 on at $t=1$, a second turns it off at $t=4$.
jump up 2 at $t=1$ jump down 2 at $t=4$

↓

2. TEMPLATE (ONE SWITCH PER JUMP)
$\boxed{\text{jump}}\cdot u(t-\boxed{\text{where}})+\boxed{\text{jump}}\cdot u(t-\boxed{\text{where}})$

↓

3. PLACE
$x(t)=2u(t-1)-2u(t-4)$

↓

4. DOWN AND BACK
$\frac{dx}{dt}=2\delta(t-1)-2\delta(t-4)$ *(each jump becomes a button)*. Integrate to return.

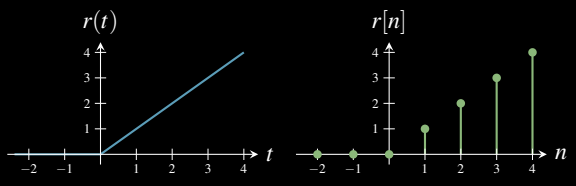
↓

5. CHECK
$t=2$: $2-0=2$ ✓ $t=5$: $2-2=0$ ✓

The ramp: the running total of the switch

$$r(t)=t\,u(t)=\begin{cases}t & t\geq 0\\ 0 & t<0\end{cases}\qquad r[n]=n\,u[n]$$

It grows without limit: neither energy nor power.



The chain: button δ $\xrightarrow{\text{running total}}$ switch u $\xrightarrow{\text{running total}}$ ramp r. Back: differentiate (CT) or first difference (DT).

Form	Formula
CT: integral of the switch	$r(t)=\int_{-\infty}^{t}u(\tau)\,d\tau$
CT: derivative of the ramp	$\frac{dr}{dt}=u(t)$
DT: running sum of the switch	$r[n]=\sum_{k=-\infty}^{n-1}u[k]$
DT: first difference	$r[n]-r[n-1]=u[n-1]$
DT: sum of shifted switches	$r[n]=\sum_{k=1}^{\infty}u[n-k]$
DT: sum of buttons	$r[n]=\sum_{k=0}^{\infty}k\,\delta[n-k]$

Trap: The DT sum stops at $n-1$: $r[1]=1$ adds only $u[0]$. Stopping at n gives $r[n+1]$.

Worked: check the running-sum form of the DT ramp

1. PIECES
In words: add up the switch, stopping one step before n. You should get $0,1,2,3$.
$u[k]$: 0 for $k<0$, 1 for $k\geq 0$

↓

2. TEMPLATE
$$r[n]=\sum_{k=-\infty}^{n-1}\boxed{u[k]}$$

↓

3. PLACE
$n=0$: nothing on yet, 0 $n=1$: $u[0]=1$ $n=3$: $u[0]+u[1]+u[2]=3$

↓

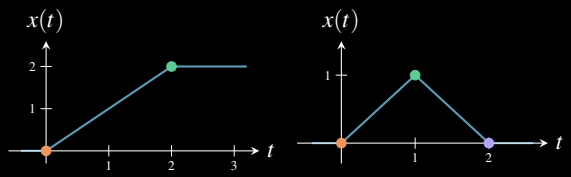
4. CHECK (THE OTHER FORM)
$r[n]-r[n-1]=u[n-1]$: $r[1]-r[0]=1=u[0]$ ✓ $r[0]-r[-1]=0=u[-1]$ ✓

Graphs from ramps and switches

Walk left to right. At each corner, add a ramp weighted by the *change in slope*. At each jump, add a switch weighted by the jump:

$$x(t)=\sum\boxed{\Delta\text{slope}}\,r\big(t-\boxed{\text{corner}}\big)+\sum\boxed{\text{jump}}\,u\big(t-\boxed{\text{where}}\big)$$

$A\,r(t-t_0)$ is 0 before t_0, then climbs with slope A.



Worked: rise $0\to 2$ on $[0,2]$, then flat (left graph)

1. PIECES (SLOPE BEFORE → AFTER)
In words: at each corner, add a ramp as big as the change in slope.
$t=0$: slope $0\to 1$, change $+1$ $t=2$: slope $1\to 0$, change -1

↓

2. TEMPLATE
$x(t)=\boxed{\Delta}\,r(t-\boxed{\text{corner}})+\boxed{\Delta}\,r(t-\boxed{\text{corner}})$

↓

3. PLACE
$x(t)=r(t)-r(t-2)$ *(the second ramp cancels the first's climb)*

↓

4. CHECK
$t=1$: 1 ✓ $t=5$: $5-3=2$, flat ✓

Worked: a triangle (right graph)

1. PIECES (SLOPE BEFORE → AFTER)
In words: up with slope 1, then down with slope -1, then flat.
$t=0$: $0\to 1$, $+1$ $t=1$: $1\to -1$, -2 $t=2$: $-1\to 0$, $+1$

↓

2. TEMPLATE
$x(t)=\boxed{\Delta}\,r(t-\boxed{\text{corner}})+\boxed{\Delta}\,r(t-\boxed{\text{corner}})+\boxed{\Delta}\,r(t-\boxed{\text{corner}})$

↓

3. PLACE
$x(t)=r(t)-2r(t-1)+r(t-2)$ *(the peak needs -2: undo $+1$, then add -1)*

↓

4. CHECK
$t=1$: 1, the peak ✓ $t=3$: $3-4+1=0$ ✓ *(slope changes sum to 0)*

Worked: (reverse) sketch $x(t)=2r(t-1)-2r(t-3)$

1. PIECES
In words: read each ramp as a corner. Its number is the change in slope.
corner at 1: slope $+2$ corner at 3: slope -2

↓

2. TEMPLATE
slope after each corner = slope before $+\boxed{\Delta}$

↓

3. PLACE
flat 0 until 1, then slope 2 until 3, then slope 0

↓

4. RESULT
a ramp from 0 at $t=1$ up to $2(2)=4$ at $t=3$, then flat at 4

↓

5. CHECK
$t=5$: $2(4)-2(2)=8-4=4$ ✓

Worked: (jump and ramps) 3 at $t=0$, then down with slope -1 to 0, then flat

1. PIECES
In words: a jump needs a switch. Each corner needs a ramp.
jump $+3$ at 0 corner at 0: slope $0\to -1$ corner at 3: slope $-1\to 0$ *($3-t=0$ at $t=3$)*

↓

2. TEMPLATE
$x(t)=\boxed{\text{jump}}\,u(t-\boxed{\text{where}})+\sum\boxed{\Delta\text{slope}}\,r(t-\boxed{\text{corner}})$

↓

3. PLACE
$x(t)=3u(t)-r(t)+r(t-3)$

↓

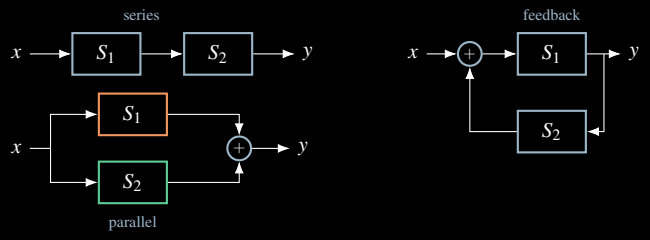
4. CHECK
$t=1$: $3-1=2$ ✓ $t=4$: $3-4+1=0$ ✓

11. systems and interconnections

THE IDEA
A **system** is a machine: a signal goes in, a different one comes out, $x \to y$. A **difference equation** is the machine's recipe: how to cook $y[n]$ from inputs, and sometimes from its own past outputs.

Three hookups:

- **Series**: an assembly line. S_1's output is S_2's input: substitute S_1 into S_2's input slot.
- **Parallel**: x feeds both, and the outputs add: $y = \boxed{y_1} + \boxed{y_2}$.
- **Feedback**: the output loops back through S_2 into the input. The machine hears its own past.



***Worked: series:* S_1*:* $w[n] = 2x[n]$*, then* S_2*:* $y[n] = w[n] - w[n-1]$**

1. PIECES
In words: S_1 doubles the input. S_2 takes the change since last step of whatever it gets.
$w[n] = 2x[n]$, so $w[n-1] = 2x[n-1]$ *(replace every n with $n-1$)*

↓

2. TEMPLATE
$y[n] = \boxed{w[n]} - \boxed{w[n-1]}$

↓

3. PLACE
$y[n] = 2x[n] - 2x[n-1]$

↓

4. CHECK
press $x = \delta$: $w = 2, 0, \ldots$, then $y = 2, -2, 0, \ldots$ formula: $y[0] = 2$, $y[1] = -2$ ✓

***Worked: parallel:* S_1*:* $2x[n]$*,* S_2*:* $x[n-1]$**

1. PIECES
In words: the input feeds both machines, and the two outputs add.
$y_1 = 2x[n]$ $y_2 = x[n-1]$

↓

2. TEMPLATE
$y = \boxed{y_1} + \boxed{y_2}$

↓

3. PLACE
$y[n] = 2x[n] + x[n-1]$

↓

4. CHECK
$x = \delta$: $y[0] = 2$, $y[1] = 1$, one stem from each branch ✓

***Worked: feedback:* $y[n] = x[n] + 0.5\,y[n-1]$*. Press the button* $x = \delta$*,* $y[-1] = 0$**

1. PIECES
In words: output now = input now + half of the last output.
$x = \delta$: 1 at $n = 0$, else 0 $y[-1] = 0$ *(at rest)*

↓

2. TEMPLATE (ONE STEP AT A TIME)
$y[n] = \boxed{x[n]} + 0.5\,\boxed{y[n-1]}$

↓

3. PLACE, RUN IT
$y[0] = 1 + 0.5(0) = 1$ *(the press)* $y[1] = 0 + 0.5(1) = 0.5$ *(only the echo)* $y[2] = 0.25$
$y[3] = 0.125$

↓

4. PATTERN
$y[n] = (0.5)^n\,u[n]$ *(each echo is half the last)*

↓

5. CHECK
$n = 3$: $(0.5)^3 = 0.125$ ✓

DT systems as difference equations

System	Difference equation	What it does
unit delay	$y[n] = x[n-1]$	one step late
gain	$y[n] = a\,x[n]$	volume knob
first difference	$y[n] = x[n] - x[n-1]$	change since last
3-point moving average	$y[n] = \frac{1}{3}\sum_{k=0}^{2} x[n-k]$	smooths
accumulator	$y[n] = \sum_{k=-\infty}^{n} x[k]$	running total
accumulator, recursive	$y[n] = y[n-1] + x[n]$	same total
bank balance	$y[n] = 1.01\,y[n-1] + x[n]$	1% interest + deposit

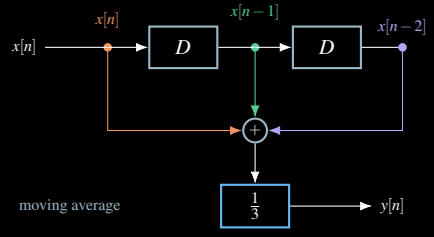
The accumulator is the running-total machine: button in, switch out. Switch in, ramp out.

Nonrecursive (FIR): inputs only. **Recursive** (IIR): also its own past outputs. It needs a start value like $y[-1] = 0$.

Block diagram parts: adder (circle) = sum of inputs. Gain (box a) = a·input. Unit delay (box D) = one-step memory, $x[n] \to x[n-1]$. A branch dot copies a signal.

Diagram ↔ equation: each wire is a piece. One D per shift, one gain per coefficient, one adder per sum.

Trap: A D that feeds $y[n]$ back into the adder means feedback: the equation has $y[n-1]$ (recursive).



Worked: (diagram to equation) read the moving-average diagram

1. PIECES (NAME EACH WIRE INTO THE ADDER)
In words: each D makes a one-step-old copy. The adder sums three copies, then the box takes a third.
$x[n]$ after one D: $x[n-1]$ after two: $x[n-2]$ gain $\frac{1}{3}$

↓

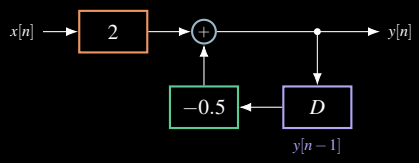
2. TEMPLATE
$y[n] = \boxed{\text{gain}}\left(\boxed{\text{wire}} + \boxed{\text{wire}} + \boxed{\text{wire}}\right)$

↓

3. PLACE
$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$

↓

4. CHECK
2 boxes D → shifts up to $n-2$ ✓ 1 adder with 3 wires → 3 terms ✓



Worked: (equation to diagram) $y[n] = 2x[n] - 0.5\,y[n-1]$

1. PIECES (ONE BLOCK PER PIECE)
In words: double the input, then subtract half of the last output.
input gain 2 $y[n-1]$: one D on the output side (feedback) its gain -0.5

↓

2. TEMPLATE
$x \to$ [gain] $\to (+) \to y$, and from y: dot $\to$ [D] $\to$ [gain] $\to$ back into $(+)$

↓

3. PLACE (DIAGRAM ABOVE)
$x \to [2] \to (+) \to y$, from y: dot $\to [D] \to [-0.5] \to (+)$

↓

4. CHECK
read it back: $y[n] = 2x[n] + (-0.5)\,y[n-1]$, the equation ✓

Worked: run the moving average: $x[0..2] = 3, 6, 9$***, others*** 0

1. PIECES
In words: slide a 3-sample window along the input, and average what is inside.
$x[0] = 3$, $x[1] = 6$, $x[2] = 9$

↓

2. TEMPLATE (A 3-SAMPLE WINDOW)
$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$

↓

3. PLACE, SLIDE THE WINDOW
$y[0] = \frac{3}{3} = 1$ $y[1] = \frac{6+3}{3} = 3$ $y[2] = \frac{9+6+3}{3} = 6$
$y[3] = \frac{9+6}{3} = 5$ $y[4] = \frac{9}{3} = 3$ then 0

↓

4. CHECK
$\sum y = 18 = \sum x$ *(each input is counted 3 times, then divided by 3)* ✓

Worked: run the accumulator $y[n] = y[n-1] + x[n]$ ***with the switch,*** $y[-1] = 0$

1. PIECES
In words: keep a running total of the input.
$x = u[n]$: 1 for $n \geq 0$ $y[-1] = 0$

↓

2. TEMPLATE
$y[n] =$ [$y[n-1]$] $+$ [$x[n]$]

↓

3. PLACE, RUN IT
$y[0] = 0 + 1 = 1$ $y[1] = 1 + 1 = 2$ $y[2] = 3$

↓

4. PATTERN
$y[n] = (n+1)u[n] = r[n+1]$ *(the running total of a switch is a ramp)*

↓

5. CHECK WITH THE SUM FORM
$y[2] = u[0] + u[1] + u[2] = 3$ ✓

12. system properties

THE IDEA
Properties are yes/no questions about a machine. Each is separate: a machine can pass one and fail another.

Property	In plain English
memoryless	It only looks at *now*.
causal	It cannot see the future. It may remember the past.
stable (BIBO)	It does not blow up: input under a ceiling ⇒ output under a ceiling.
invertible	You can undo it: different inputs give different outputs.
time invariant	It ignores the clock: delay in ⇒ same output, delayed.
linear	Scale in → scale out, add in → add out: $c_1x_1 + c_2x_2 \to c_1y_1 + c_2y_2$.

Quick red flags

Trap: Linearity: put in $x = 0$. A linear machine gives $y = 0$. The $+3$ in $2x + 3$ fails.
Trap: Time invariance: an n or t *outside* $x(\cdot)$ ($nx[n]$) reads the clock. Scaling or flipping inside ($x(2t)$, $x(-t)$) also fails.
Trap: Causality: an advance like $x[n+1]$. And $y(t) = x(-t)$: at $t = -1$ it needs $x(1)$.
Trap: Stability: an n multiplying x, or a sum with no end, can blow up.

Memory and causality: read the time stamps

Every $x[\cdot]$ on the right carries a time stamp. Color each one: now (n), past ($n-1, \ldots$), future ($n+1, \ldots$).

memoryless $\iff$ every stamp is [now] causal $\iff$ no stamp is [future]

Worked: memory and causality of $y[n] = x[n] + x[n-1]$

1. PIECES (COLOR THE STAMPS)
In words: add the input now to the input one step ago.
$x[n]$: now $x[n-1]$: past

↓

2. TEMPLATES
memoryless: every stamp [now] causal: no [future] stamp

↓

3. PLACE
a past stamp ⇒ has memory no future stamp ⇒ causal

↓

4. CHECK
$n = 5$ uses $x[5]$ and $x[4]$: the past, never $x[6]$ ✓

Worked: invertibility of $y = x^2$

1. PIECES
In words: square the input now. Can you get the input back from the output?
output y

↓

2. TEMPLATE
solve for x: exactly one answer ⇒ invertible

↓

3. PLACE
$x = \pm\sqrt{y}$: two answers ⇒ not invertible

↓

4. CHECK
inputs 1 and -1 both give 1, so the sign is lost ✓

Quick example	Work → answer
$y[n] = x[n+1]$	a future stamp: memory, not causal
$y[n] = x[n]^2$	every stamp now: memoryless, causal
$y = 2x$	$x = \frac{y}{2}$, one answer: invertible

Time invariance: two paths, then compare

Path 1: delay the input, then run the machine — change only the inside of x.
Path 2: run the machine, then delay the output — change *every* n, including any lone n. Same result ⇒ time invariant.

Worked: is $y[n] = nx[n]$ time invariant?

1. PIECES
In words: multiply the input now by the time n.
the inside of x: n a lone n outside x delay n_0

↓

2. TEMPLATE
Path 1: replace only $\boxed{\text{the inside}}$ Path 2: replace every n, the $\boxed{\text{lone } n}$ too

↓

3. PLACE: PATH 1 (DELAY, THEN MACHINE)
delayed input $x[n-n_0]$ goes in the slot: $nx[n-n_0]$ *(the machine still multiplies by its own clock n)*

↓

4. PLACE: PATH 2 (MACHINE, THEN DELAY)
run first: $nx[n]$. Delay the whole output: $(n-n_0)x[n-n_0]$

↓

5. COMPARE
$nx[n-n_0] \neq (n-n_0)x[n-n_0] \Rightarrow$ time varying *(the lone n reads the clock)*

↓

6. CHECK (RED FLAG)
an n outside $x(\cdot)$ is the red flag, and it fails ✓

Quick example	Work → answer
$x(2t)$	Path 1: $x(2t-t_0)$ Path 2: $x(2(t-t_0))$: time varying
$x[n-1]$	Path 1: $x[n-1-n_0]$ Path 2: $x[n-n_0-1]$: time invariant

Linearity: combine first, or combine after?

Path 1: mix the inputs ($c_1x_1 + c_2x_2$), then run the machine. **Path 2**: run each input alone, then mix the outputs with the same knobs. Linear means the two paths agree.
The knobs c_1, c_2 are any two numbers. Use c_1, c_2, not a, b: a system can have its own constants named a or b.

Worked: is $y[n] = \dfrac{x[n]}{n}$, $n \neq 0$, linear?

1. PIECES
In words: take the input *now*, and divide it by the time n.
inputs x_1, x_2 knobs c_1, c_2

↓

2. TEMPLATE
Path 1: $S\{c_1\boxed{x_1} + c_2\boxed{x_2}\}$ Path 2: $c_1 S\{\boxed{x_1}\} + c_2 S\{\boxed{x_2}\}$

↓

3. PLACE: PATH 1 (MIX, THEN MACHINE)
mixed input $w[n] = c_1x_1[n] + c_2x_2[n]$. Write w wherever the recipe says x:
$\dfrac{w[n]}{n} = \dfrac{c_1x_1[n] + c_2x_2[n]}{n} = \dfrac{c_1x_1[n]}{n} + \dfrac{c_2x_2[n]}{n}$ *(one fraction over a common denominator splits into two)*

↓

4. PLACE: PATH 2 (MACHINE, THEN MIX)
run each alone: $y_1[n] = \dfrac{x_1[n]}{n}$, $y_2[n] = \dfrac{x_2[n]}{n}$
mix with the same knobs: $c_1y_1[n] + c_2y_2[n] = \dfrac{c_1x_1[n]}{n} + \dfrac{c_2x_2[n]}{n}$

↓

5. COMPARE
Path 1 = Path 2 for every c_1, $c_2 \Rightarrow$ linear *(but time varying: the lone n reads the clock. Separate properties)*

↓

6. CHECK (ZERO TEST)
$x = 0$: $y = \dfrac{0}{n} = 0$ ✓ *(a linear system must pass. Passing alone does not prove linear)*

Quick example	Work → answer
$y = 2x[n] + 3$	zero test: $2(0) + 3 = 3 \neq 0 \Rightarrow$ not linear *(incrementally linear)*
$y = nx[n]$	same steps as above, with $\times n$ for $\div n$: linear

Stability with a ceiling B

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BIBO: bounded in, bounded out. "Bounded" means a ceiling: $|x[n]| \leq B < \infty$. Stable means the output has a ceiling too. This is Lesson 1 again: is the signal's size finite?

To show stable: take $|\cdot|$ of the equation, split with $|a+b| \leq |a| + |b|$ and $|cx| = |c||x|$, then replace each $|x[\cdot]|$ (shifted ones too) with B:

$$|y| \leq \boxed{|c_1|}B + \boxed{|c_2|}B + \cdots$$

A finite number (no n, no endless sum) means stable. **To show unstable**: one capped input (like $x = 1$ or $u[n]$) with an unbounded output.
Trap: $|nx[n]| \leq |n|B$ grows with n. The accumulator adds another B every step.

Worked: is $y[n] = 2x[n] + x[n-1]$ stable?

1. PIECES
In words: twice the input now plus the input one step ago. Cap the input at B.
coefficients 2, 1

↓

2. TEMPLATE
$|y| \leq \boxed{|c_1|}B + \boxed{|c_2|}B$

↓

3. PLACE
$|y| \leq 2B + 1B = 3B$ *(split the absolute value, then replace each $|x|$ with B)*

↓

4. LABEL
$3B$ is a finite number with no n in it $\Rightarrow$ stable

↓

5. CHECK (SECTION 13)
$\sum|h| = 2 + 1 = 3 < \infty$ ✓

Worked: is $y[n] = nx[n]$ stable?

1. PIECES
In words: multiply the input by the time n. Try the simplest capped input.
test input $x = 1$ for all n *(capped at $B = 1$)*

↓

2. TEMPLATE
one capped input with an unbounded output $\Rightarrow$ not stable

↓

3. PLACE
$y[n] = n \cdot 1 = n$ *(it keeps growing)*

↓

4. LABEL
a capped input gave an output with no ceiling $\Rightarrow$ not stable

↓

5. CHECK (RED FLAG)
an n multiplying x is the stability red flag ✓

Quick example	Work → answer
moving average	$\leq \frac{1}{3}(B+B+B) = B$: stable
$y = e^{x[n]}$	$-B \leq x \leq B \Rightarrow |y| \leq e^B$: stable
accumulator	input $u[n]$ gives $y = n+1$: not stable
$y = 0.5y[n-1] + x$	unroll: $|y| \leq B\frac{1}{1-0.5} = 2B$: stable *(with 1.01: unstable)*

Properties of common DT systems

System	Mless	Causal	Linear	TI	Stable
$x[n-1]$	no	yes	yes	yes	yes
$x[n] - x[n-1]$	no	yes	yes	yes	yes
moving average	no	yes	yes	yes	yes
accumulator	no	yes	yes	yes	no
$nx[n]$	yes	yes	yes	no	no
$x[n]^2$	yes	yes	no	yes	yes
$2x[n] + 3$	yes	yes	no	yes	yes
$x[n+1]$	no	no	yes	yes	yes
$x[2n]$	no	no	yes	no	yes
$x[-n]$	no	no	yes	no	yes
$0.5y[n-1] + x[n]$	no	yes	yes	yes	yes

Mless = memoryless, TI = time invariant. The recursive system starts at rest.

Trap: The properties are separate: linear but time varying ($nx[n]$), stable but not causal ($x[n+1]$).

13. the convolution sum

THE IDEA
Section 10 showed that every DT input is a stack of buttons: $x[n] = \sum_k x[k]\,\delta[n-k]$. Press the button once and an LTI machine answers with its **impulse response** $h[n]$, its "echo." Each button in the stack makes its own echo, scaled by its height and shifted to its time. The output is the sum of all the echoes. That sum is the **convolution sum**.

Why LTI makes it work, one property per line:

Input	Output
$\delta[n]$	$h[n]$ *(definition of h)*
$\delta[n-k]$	$h[n-k]$ *(time invariant)*
$x[k]\,\delta[n-k]$	$x[k]\,h[n-k]$ *(linear: scale)*
$\sum_k x[k]\,\delta[n-k] = x[n]$	$\sum_k x[k]\,h[n-k] = y[n]$ *(linear: add)*

Formula	In words
$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$	(output at n) = sum over every input time k of (input at k)(echo, $n-k$ steps later)
$y[n] = x[n] * h[n]$	"x convolved with h"
$y[n] = \sum_k h[k]\,x[n-k]$	same result, roles swapped: $x*h = h*x$
$h[n]$ = output for $x = \delta$, at rest	the **impulse response**: it describes the whole LTI system

Colors in this section: the impulse response h is always green. The input stems are orange, blue, yellow, one color per stem. Purple is the output time n.

Trap: k is the dummy variable that sweeps. n stays fixed during one sum. Each n is a new sum.

Trap: $h[n-k]$, not $h[k-n]$. And $y[n] \neq x[n]\,h[n]$: convolution is not a product of the two signals.

Trap: Only for LTI systems that start at rest. $y = x^2$ or $y = nx[n]$ has no impulse-response formula.

The impulse response describes the system

Formula	In words
causal $\iff h[n] = 0$ for $n < 0$	no echo before the press
BIBO stable $\iff \sum_n \lvert h[n]\rvert < \infty$	the echoes add to a finite size
memoryless $\iff h[n] = c\,\delta[n]$	the echo is only at $n = 0$
FIR: h has finitely many stems	$y[n] = \sum_k h[k]\,x[n-k]$, a nonrecursive equation
IIR: h goes on forever	usually from a recursive (feedback) equation
series h_1 then h_2	one system with $h = h_1 * h_2$
parallel h_1 and h_2	one system with $h = h_1 + h_2$

Quick example	Work → answer
unit delay $y = x[n-1]$	$h = \delta[n-1]$
first difference	$h = \delta[n] - \delta[n-1]$
3-point moving average	$h = \frac{1}{3}\left(\delta[n] + \delta[n-1] + \delta[n-2]\right)$
accumulator	$h = u[n]$: $\sum \lvert h\rvert = \infty$, not stable
$y = 0.5y[n-1] + x$	$h = (0.5)^n u[n]$ *(section 11)*: $\sum = 2$, stable

Which way? — system, impulse response, output

Given → find	Do this
nonrecursive equation → h	read the coefficients: $y = \sum b_k x[n-k]$ gives $h[k] = b_k$
recursive equation → h	put in $x = \delta$, start at rest, run it step by step
h → equation	$y[n] = \sum_k h[k]\,x[n-k]$, one term per stem
h → causal, stable, memoryless	the h tests above
x and h → y	convolve: table, flip and slide, or formula
output for $x = \delta$ → h	that output *is* h

Worked: find h for $y[n] = x[n] + 2x[n-1] - x[n-2]$

1. PIECES
In words: press the button and see what comes out. Each input term becomes one stem of h.
coefficients $1,\ 2,\ -1$ at delays $0,\ 1,\ 2$

↓

2. TEMPLATE
$h[n] = \boxed{b_0}\,\delta[n-\boxed{0}] + \boxed{b_1}\,\delta[n-\boxed{1}] + \boxed{b_2}\,\delta[n-\boxed{2}]$

↓

3. PLACE
$h[n] = 1\,\delta[n] + 2\,\delta[n-1] - 1\,\delta[n-2]$ *(an FIR system: 3 stems)*

↓

4. CHECK
put in $x = \delta$ by hand: $y[1] = \delta[1] + 2\delta[0] - \delta[-1] = 2$, the second stem ✓

Worked: find h for $y[n] = 0.8\,y[n-1] + x[n]$

1. PIECES
In words: press the button once, start at rest, and run the feedback loop.
input $x = \delta$ start $y[-1] = 0$ feedback gain 0.8

↓

2. TEMPLATE
$h[n] = \boxed{0.8}\,h[n-1] + \boxed{\delta[n]}$

↓

3. PLACE, RUN IT
$h[0] = 0.8(0) + 1 = 1$ $h[1] = 0.8(1) + 0 = 0.8$ $h[2] = 0.64$ *(each echo is 0.8 of the last)*

↓

4. PATTERN
$h[n] = (0.8)^n u[n]$ *(an IIR system: it never stops)*

↓

5. CHECK
$\sum (0.8)^n = \frac{1}{1-0.8} = 5 < \infty$: stable ✓ $h[n] = 0$ for $n < 0$: causal ✓

Worked: (reverse) $h[0] = 3$, $h[1] = -1$, else 0. Write the system and its properties

1. PIECES
In words: each stem of h becomes one input term.
$h[0] = 3$, $h[1] = -1$

↓

2. TEMPLATE
$y[n] = \boxed{h[0]}\,x[n] + \boxed{h[1]}\,x[n-1]$

↓

3. PLACE
$y[n] = 3x[n] - 1\,x[n-1]$

↓

4. PROPERTIES
causal *(no stem before 0)* stable ($\sum \lvert h\rvert = 4$) not memoryless *(a stem at 1)*

↓

5. CHECK
$x = \delta$: $y[0] = 3$, $y[1] = -1$, the stems of h ✓

Method 1: the table (one shifted echo per input stem)

THE IDEA
Each input stem $x[k]$ launches its own copy of h, scaled by $x[k]$ and starting at $n = k$. Write one row per stem, then add each column. It is long multiplication without carrying.

Formula	In words
row for stem k	$x[k] \cdot h$, shifted to start at $n = k$
$y[n]$	sum of column n
x on $[a,b]$, h on $[c,d]$	y on $[a+c,\ b+d]$
length of y	$L_x + L_h - 1$
$\sum_n y[n] = \left(\sum_n x[n]\right)\left(\sum_n h[n]\right)$	quick check of the whole answer

Worked: $x[-1]=2$, $x[0]=-1$, $x[1]=3$ ***and*** $h[0]=1$, $h[1]=2$

1. PIECES
In words: three input stems, so three echoes. Each echo is h scaled by its stem and started at its time.
input stems 2 at -1, -1 at 0, 3 at 1 echo $h=1,2$ from 0

↓

2. TEMPLATE (ONE ROW PER STEM)
row k: $\boxed{x[k]}\cdot(\boxed{1},\boxed{2})$, starting at column $n=k$
range $[-1+0,\ 1+1]=[-1,2]$, length $3+2-1=4$

↓

3. PLACE

n	-1	0	1	2
$2\cdot h$	2	4		
$-1\cdot h$		-1	-2	
$3\cdot h$			3	6
$y[n]$	2	3	1	6

↓

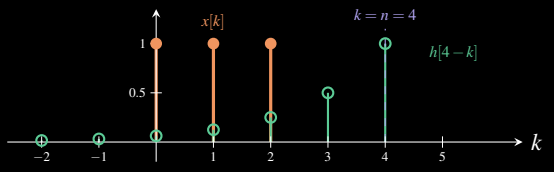
4. CHECK
$\sum y=12$ and $(\sum x)(\sum h)=(4)(3)=12$ ✓ 4 stems on $[-1,2]$ ✓

Method 2: flip and slide (one output time at a time)

THE IDEA
Fix one output time n. Draw $x[k]$. Flip h to get $h[-k]$, then slide it right by n to get $h[n-k]$. Multiply the two stem by stem where they overlap and add. Then move to the next n.

1. Draw $x[k]$ on a k axis.
2. Flip: $h[k]\to h[-k]$. Slide: $h[-k]\to h[n-k]$. Its first stem sits at $k=n$, the rest trail to the *left*.
3. Multiply where both are nonzero. Add. That is $y[n]$.
4. Split n into regions: no overlap ($y=0$), partial overlap, full overlap. Each region gets one formula.

Trap: After the flip, $h[n-k]$ is *backwards*: $h[0]$ sits at $k=n$, and $h[1]$ sits at $k=n-1$.



The picture is the next example at $n=4$: the pulse $x[k]$ and the flipped echo $h[4-k]$. The overlap is $k=0,1,2$.

Worked: $x[n]=u[n]-u[n-3]$ ***(on at*** $0,1,2$***)*** ***and*** $h[n]=(0.5)^n u[n]$

1. PIECES
In words: a 3-sample pulse meets an echo that halves each step. Slide the flipped echo across and watch the overlap grow, then stay full.
$x[k]=1$ for $k=0,1,2$ $h[n-k]=(0.5)^{n-k}$ for $k\le n$ regions: $n<0$, $0\le n\le 2$, $n\ge 3$

↓

2. TEMPLATE
$$y[\boxed{n}]=\sum_{k\in\text{overlap}}\boxed{x[k]}\,\boxed{h[n-k]}$$

↓

3. PLACE: NO OVERLAP, $n<0$
the flipped echo is all left of 0 $\Rightarrow$ $y[n]=0$

↓

4. PLACE: PARTIAL OVERLAP, $0\le n\le 2$ ($k=0..n$)
$$y[n]=\sum_{k=0}^{n}1\cdot(0.5)^{n-k}=\sum_{m=0}^{n}(0.5)^m=2\left(1-(0.5)^{n+1}\right)$$ *(rename $m=n-k$, then the finite geometric sum)*

↓

5. PLACE: FULL OVERLAP, $n\ge 3$ ($k=0,1,2$)
$y[n]=(0.5)^n+(0.5)^{n-1}+(0.5)^{n-2}=(0.5)^n(1+2+4)=7(0.5)^n$ *(pull out $(0.5)^n$)*

↓

6. CHECK
the regions meet: $n=2$ gives $2(1-0.125)=1.75$ and $7(0.25)=1.75$ ✓
$n=4$ (picture): $0.0625+0.125+0.25=0.4375=7(0.5)^4$ ✓

Method 3: formula (infinite signals)

THE IDEA
When both signals are formulas, put them in the sum. The switches $u[k]$ and $u[n-k]$ cut the sum down to $0\le k\le n$. Then use the geometric sum.

Formula	In words
$u[k]\,u[n-k]$	1 for $0\le k\le n$, else 0: the sum runs $k=0..n$, and $y=0$ for $n<0$
$\sum_{k=0}^{n} r^k=\dfrac{1-r^{n+1}}{1-r}$	finite geometric sum, $r\neq 1$
$a^n u[n]*b^n u[n]=\dfrac{a^{n+1}-b^{n+1}}{a-b}\,u[n]$	two exponentials, $a\neq b$
$a^n u[n]*a^n u[n]=(n+1)\,a^n u[n]$	same base

Quick example	Work → answer
$x*\delta$	x: the button changes nothing
$x*\delta[n-3]$	$x[n-3]$: a delay by 3
$\delta[n-1]*\delta[n-4]$	$\delta[n-5]$: delays add
$u*u$	$\sum_{k=0}^{n}1=(n+1)u[n]=r[n+1]$ *(the accumulator of a switch, section 11)*
$u[n]*(0.5)^n u[n]$	$\sum_{k=0}^{n}(0.5)^k=2\left(1-(0.5)^{n+1}\right)u[n]$

Worked: $x[n]=(0.5)^n u[n]$ ***and*** $h[n]=(0.25)^n u[n]$

1. PIECES
In words: both signals start at 0 and shrink, so the switches limit the sum to $k=0..n$.
$x[k]=(0.5)^k u[k]$ $h[n-k]=(0.25)^{n-k}u[n-k]$ switches: $0\le k\le n$

↓

2. TEMPLATE
$$y[\boxed{n}]=\sum_{k=0}^{\boxed{n}}\boxed{x[k]}\,\boxed{h[n-k]}$$

↓

3. PLACE
$$y[n]=\sum_{k=0}^{n}(0.5)^k\,(0.25)^{n-k}=(0.25)^n\sum_{k=0}^{n}2^k$$ *(pull out $(0.25)^n$, and $\frac{0.5}{0.25}=2$)*

↓

4. GEOMETRIC SUM
$\sum_{k=0}^{n}2^k=2^{n+1}-1$ $\Rightarrow$ $y=(0.25)^n\left(2^{n+1}-1\right)=\left(2(0.5)^n-(0.25)^n\right)u[n]$

↓

5. CHECK
$y[0]=x[0]h[0]=1$, and $2-1=1$ ✓ $y[1]=0.25+0.5=0.75$, and $1-0.25=0.75$ ✓

Which method?

If the signals are …	Use
both short lists of stems	the table (Method 1)
one short pulse, one long formula	flip and slide, by regions (Method 2)
both formulas with switches	the formula (Method 3)
one of them is $\delta[n-n_0]$	shift the other by n_0